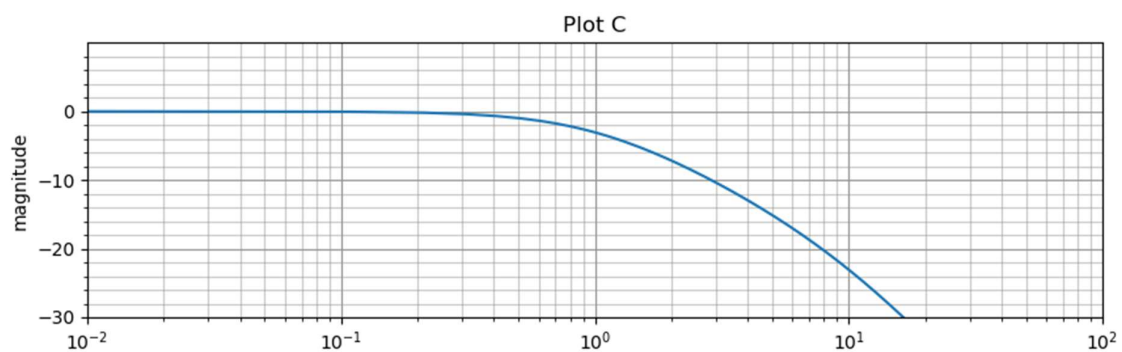
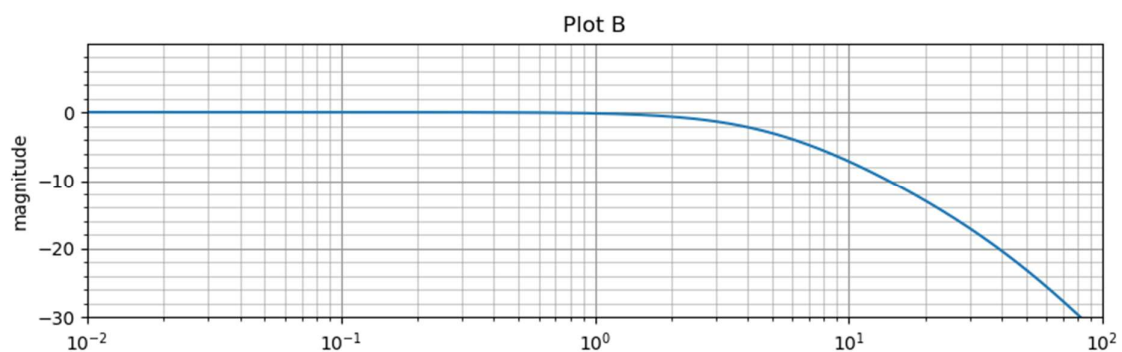
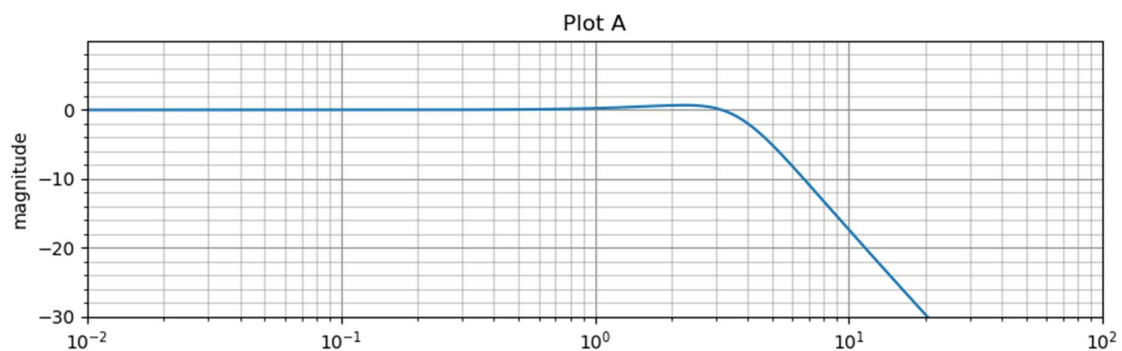
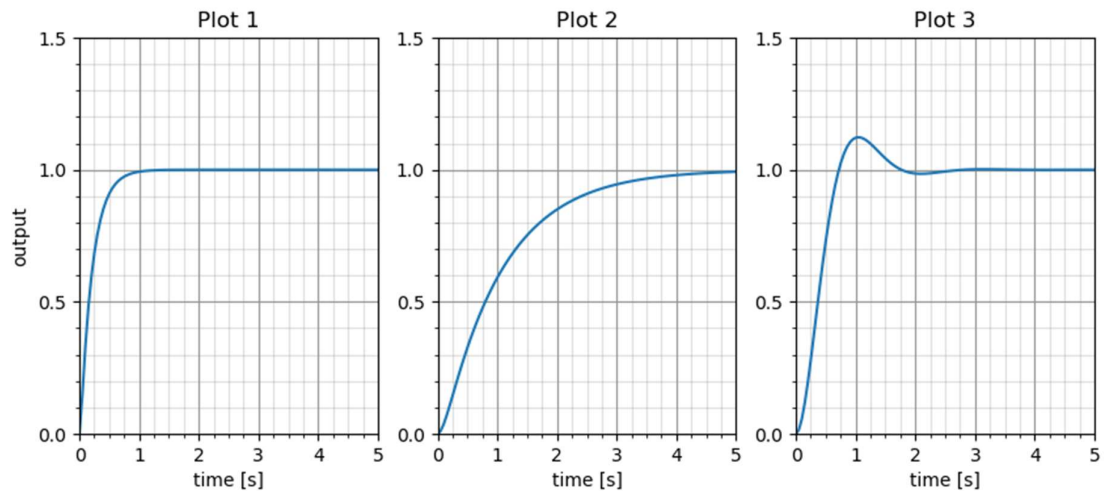


Tutorial 3: System Identification – Frequency Domain

Question 1: time and frequency responses

Each of the following plots 1-3 represents the unit step response for a second-order system, while the plots A-C give the magnitudes of their frequency responses. Match the corresponding plots, briefly justifying your selections.



Solutions:

1 = B, 2 = C, 3 = A

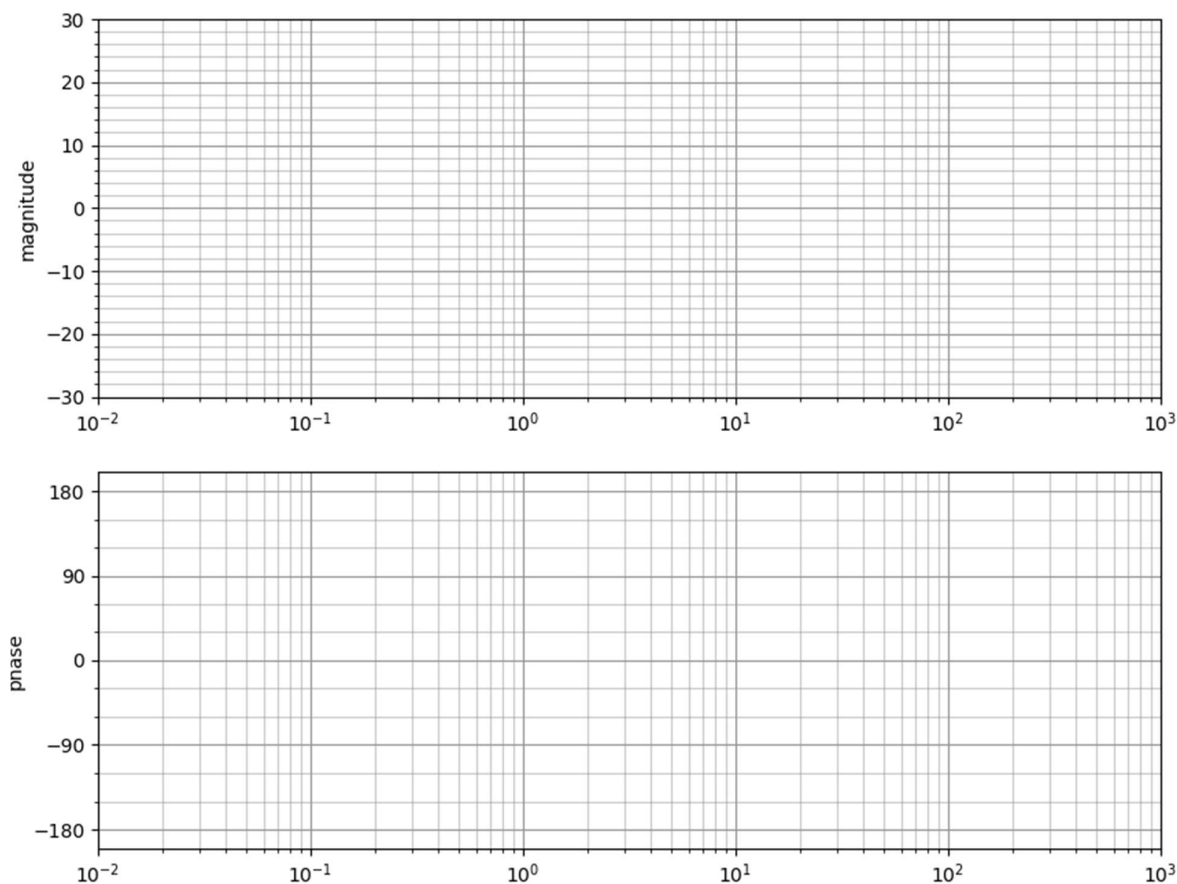
Plot 1 and 2 show systems with two real poles, as there is no overshoot. They therefore can't have the frequency response shown in A, as the slight resonant peak in that plot can only occur with complex poles.

Only the response in 3 has overshoot, so this must match with A.

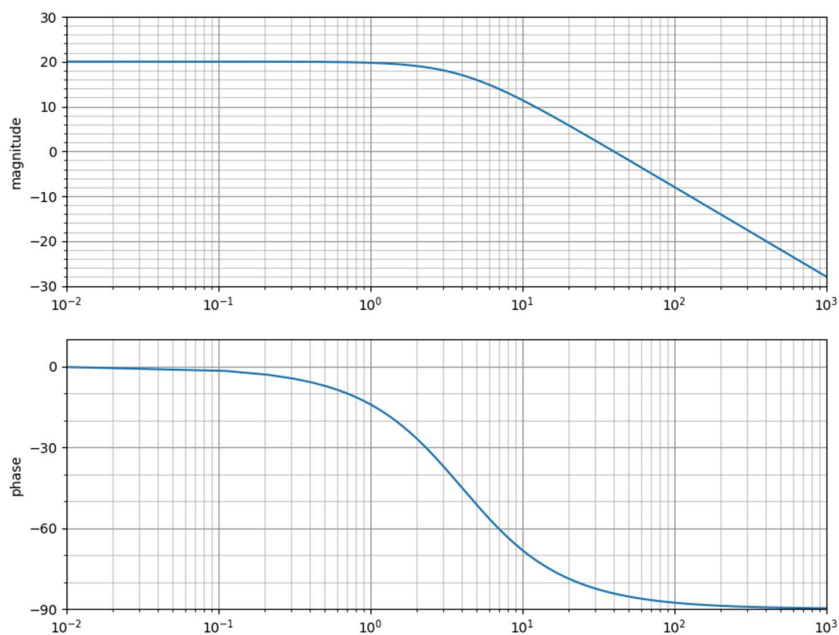
Both B and C could match with systems with real poles, so we must use the corner frequency to determine which matches each step response. The system in plot 1 is faster, and therefore will have a corner frequency, so it goes with B and the slower system in plot 2 goes with C.

Question 2: first-order frequency responses

- a) Give a rough* sketch of the frequency response of the system $G(s) = \frac{20}{s+10}$ on the Bode plot below. Check your work by selecting five frequencies spanning a sensible range and calculating the magnitude and phase of the frequency response for these values. Show the resulting points on your plot. (rough* = asymptotic. We want to see the low and high frequency asymptotes, and the corner frequency)



- b) Determine the transfer function of the first-order system with the following frequency response:



Solutions:

a) **Magnitude**

The pole of the system is at -10 , so the corner frequency will be at 10 rad/s , where the low- and high-frequency asymptotes intersect. The gain of the transfer function is $\lim_{s \rightarrow 0} \frac{20}{s+10} = 2 = 6 \text{ dB}$. This gives the value of the low frequency asymptote. The high-frequency asymptote will roll off at 20 dB/decade from the corner frequency, as there is just one pole.

Phase

Based on the standard first-order frequency response, the phase will transition from 0 degrees at high frequencies to -90 degrees at high frequencies. It will reach -45 degrees at the corner frequency, and we can use a decade up and down from this value to approximate the transition.

Test points

You can calculate these values by substituting different values of $j\omega$ into the transfer function using the complex function on your calculator. I selected the corner frequency and the frequencies 1 and 2 decades up and down from this:

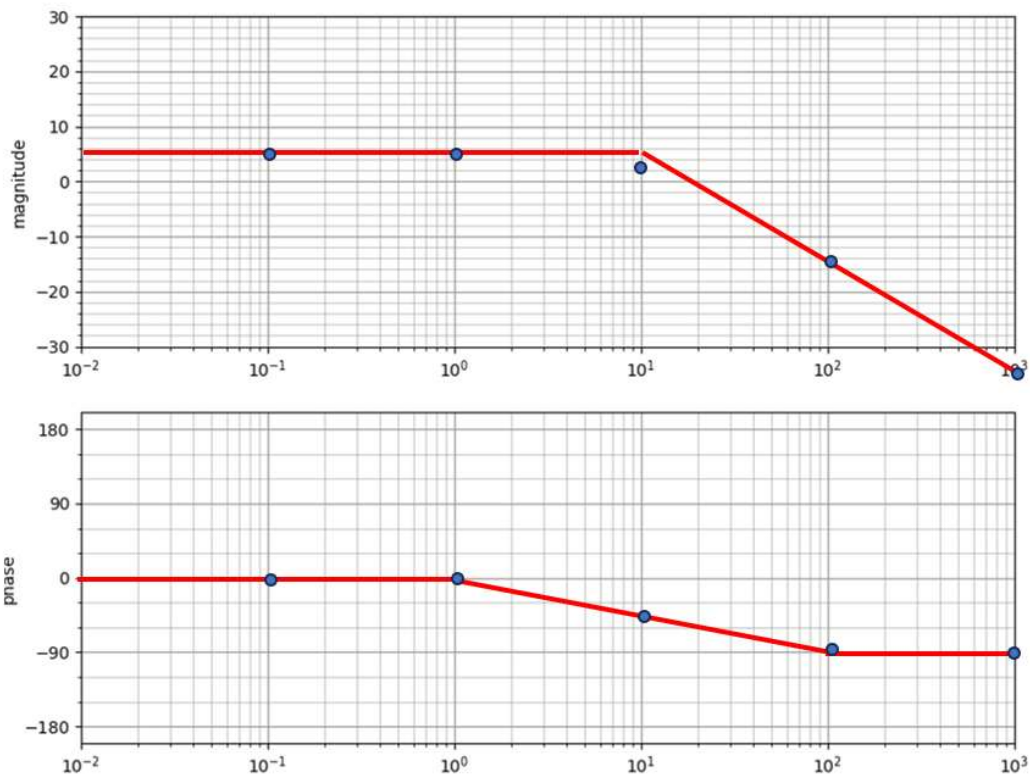
$$1. \frac{20}{j(0.1)+10} = 2 - 0.02j = 6 \text{ dB}, -0.5^\circ$$

$$2. \frac{20}{j(1)+10} = 2 - 0.2j = 6 \text{ dB}, -5^\circ$$

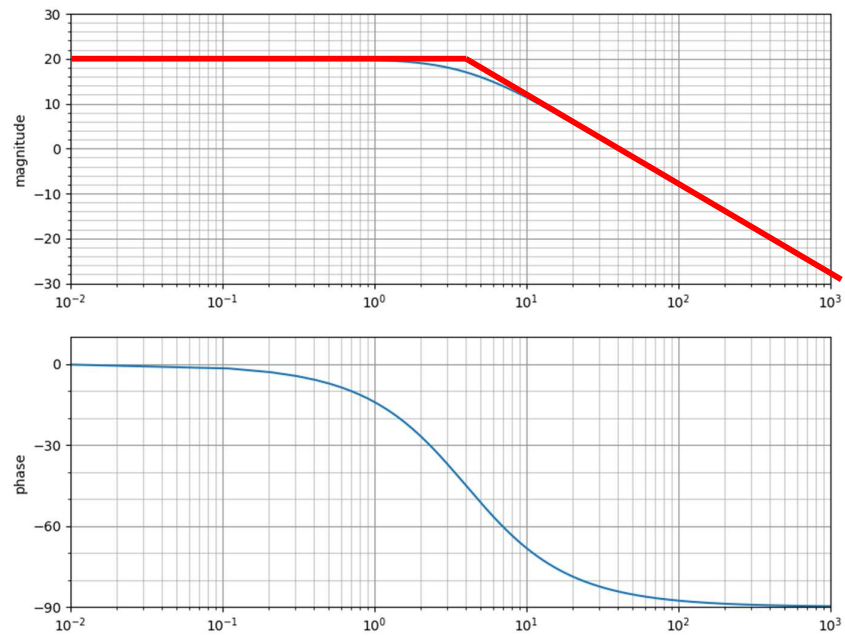
$$3. \frac{20}{j(10)+10} = 1 - 1j = 3 \text{ dB}, 45^\circ$$

$$4. \frac{20}{j(100)+10} = 0.02 - 0.2j = -14 \text{ dB}, -84^\circ$$

$$5. \frac{20}{j(1000)+10} = 0 - 0.2j = -34 \text{ dB}, -90^\circ$$



b) It is helpful to draw the asymptotes on the plot:



The asymptotes intersect at 1,4 rad/s, which is also where there is a phase of -45° , so this must be the corner frequency. This indicates that the pole is at $s = -1,4$.

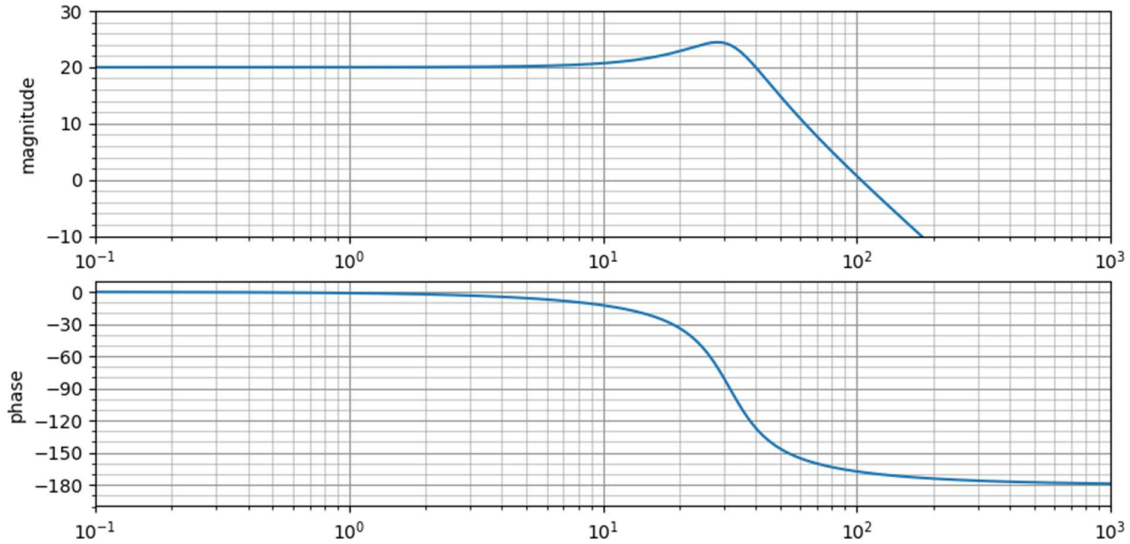
The low-frequency gain of the system is $A = 20$ dB, or 10, so the transfer function is $\frac{100}{\frac{1}{1,4}s+1}$ or $\frac{14}{s+1,4}$

Question 3: second-order frequency responses

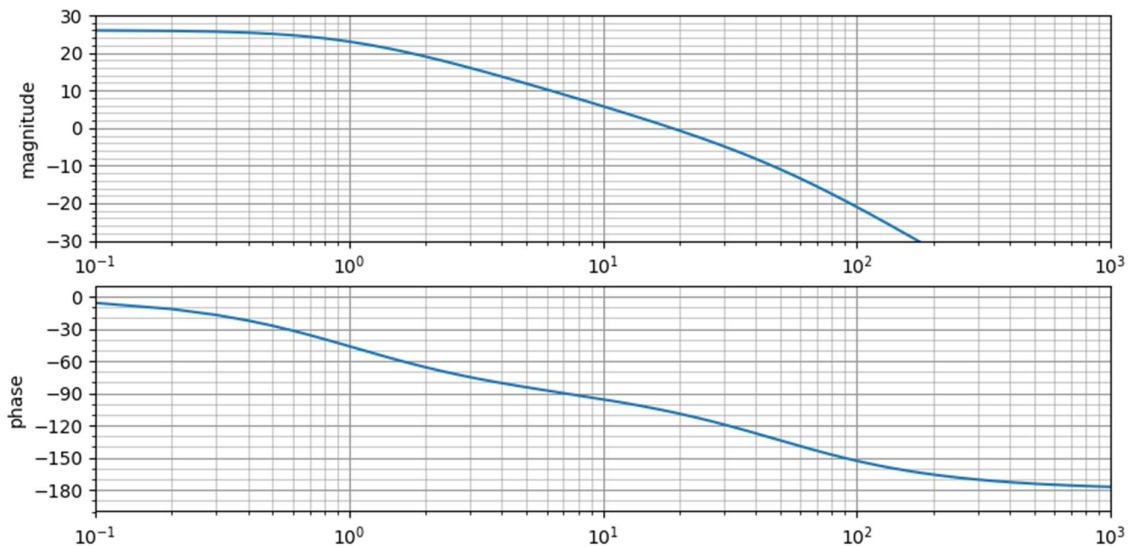
The Bode plots below were produced by second-order systems. Based on these responses,

- Do you expect the transfer function to have two real poles, or a complex pole pair? Explain.
- Determine the transfer function of the system.

Plot A



Plot B



Solutions:

- The magnitude plot has a resonant peak so the system is underdamped/ has complex poles.
 - The system is underdamped, so the easiest system identification approach will be to try to find the gain, natural frequency and damping ratio.
The low-frequency gain of the system is $A = 20 \text{ dB} = 10$.
The resonant peak and crossing through -90° happen at just above 30 rad/s, so we can estimate that the natural frequency $\omega_n = 31$.
The value at the resonant peak is around 4 dB = 1.58 above the low frequency gain.

So $\frac{1}{2\zeta} = 1.58, \zeta = 0.32$.

This gives the transfer function $\frac{9610}{s^2 + 19.6s + 96}$

The actual transfer function used to generate this plot was $\frac{10000}{s^2 + 20 + 1000}$, which we have not recreated perfectly, but the values are close enough. In a test or exam, we can't expect you to eyeball values perfectly so these will be marked with a wide tolerance.

b)

- i) The system is overdamped/ has two real poles.
- ii) This time we are going to try to find the values of the two real poles and the gain.

The gain is around 26 dB = 20.

For the poles, it is helpful to draw asymptotes on the plot. They intersect at around 1 and 50 rad/s, which is also where the phase plot crosses -45° and -135° . This gives poles at -1 and -50.

The overall transfer function is then $\frac{1000}{(s+1)(s+50)}$

